



Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n)
```

```
{
```

```
    int i= 1;
```

```
    int s =1;
```

```
    while(s <= n)
```

```
    {
```

```
        i++;
```

```
        s += i;
```

```
    }
```

```
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

For example:

Input	Result
9	12

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  void function(int n) {
4      int i = 1;
5      int s = 1;
6      int count = 2;
7      while(s <= n) {
8          count++;
9          i++;
10         count++;
11         s += i;
12         count++;
13     }
14     count++;
15     printf("%d", count);
16 }
17
18 int main() {
19     int n;
20     scanf("%d", &n);
21     function(n);
22     return 0;
23 }
```

Check

	Input	Expected	Got	
✓	9	12	12	✓
✓	4	9	9	✓

Passed all tests! ✓

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